

7824 Zero Road, second floor

M-6 HVAC plan · packet for the hvac contractor

Supply and return ductwork in the floor trusses from AH-1 below, floor registers and returns, zoning. Read it with the hydronic pack, sheets M-1 to M-5 (packets/M-1-to-M-5-hydronic-pack.pdf): the radiant floor heats this floor and shares the truss bays with these ducts.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example M-6 G7).

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Notes

1. AH-1, the 6-ton hydronic air handler (2,400 CFM) on hydronic sheet M-3, sits on the floor below at X 5', Y 7', directly under the stair landing. It serves both floors through the D-UP and D-DN dampers on sheet M-5; this sheet ducts the second floor only, at its share of about 1,325 CFM (0.4 CFM per sq ft). The first floor gets its own trunk off the handler later and is not routed through this floor. Supply riser 18" × 10" and return riser 20" × 20" come up through the floor into the 24" open-web truss space.
2. Heat on this floor is the radiant floor on hydronic sheet M-1: PEX in every truss bay, zones A, B and C on thermostats T1 to T3. AH-1 heats the floor below (hydronic sheet M-4) and runs heat-only until the Phase 2 heat pump HP-1 brings cooling.
3. Until the first floor is ducted, do not push the full 2,400 CFM through this floor: set the ECM blower to about 1,350 CFM (or fit a barometric bypass at the handler). The ducts here are sized for 1,325 CFM and would be noisy and over-pressured at 2,400.
4. Every duct runs inside the floor trusses: rectangular trunks ST1 to ST5 at the truss bottom chord (the 18" × 10" main trunk leaves room under it for the drain mains on P-1), 5" and 6" round branches to floor registers set along the exterior walls under the windows. Registers total 1,325 CFM at about 0.08" per 100' friction; the HVAC contractor balances the rooms with a Manual J / D.
5. No floor grilles anywhere (wheelchair). Returns are wall grilles at the baseboard, 4" to 12" AFF: R1 on the living face of wall 27, R2 on the bedroom face of wall 19, R3 on the hall face of wall 7, R4 on the den face of wall 17, R5 on the dining face of wall 32. A lined stud bay passes about 250 CFM in a 2x6-plate wall and 180 in a 2x4 wall, so each return spans the number of bays listed (one 14×8 grille per bay, side by side) and the bays join in a plenum box in the truss space before the rectangular duct runs back to the return riser. Walls 17, 19, 27 and 32 are 2x6-plate staggered walls; wall 7 is 2x4.
6. Supply registers are floor registers because the air comes up through the floor; they are flush, heavy-gauge steel, screwed to the floor, 4" wide with no raised lip, so a wheel crosses them like the floor. If even that is unwelcome, say so: the alternative is toe-kick registers under the island and vanities plus baseboard supplies in the 2x6 walls, at the cost of less even heat under the windows.
7. Two air zones with motorised dampers at the branch trunks: zone 1 public rooms and the master suite, zone 2 the bedroom wing. Thermostats T6 on the office wall facing the hall and T7 in the wing hall, 48" AFF (T1 to T5 are the hydronic pack's, sheet M-5). See the zoning open item.
8. Bath exhaust, the dryer and the cooktop downdraft are on P-1 (exhaust for coordination). The ERV in the mech room ducts fresh air into the return trunk at the riser and exhausts from the baths; its 6" ducts also run in the floor.
9. Floor registers: steel 4" × 10" / 12" / 14" with opposed-blade dampers, flush, no raised lips (wheelchair). Keep registers 12" clear of pocket-door cavities and the drains on P-1; the drain mains at Y 21'-6" and X 40' cross the trunks and win the lower position.
10. Insulate supply trunks in the truss space (R-6 wrap); seal every joint with mastic; test to 4 CFM per 100 sq ft.

Open items

- Equipment: AH-1 serves both floors from its hydronic coil (B-1 now, HP-1 later; hydronic sheet M-3). Confirm its external static (0.5" assumed) and that its ECM blower can run at this floor's 1,325 CFM until the first floor is connected. Floor-to-floor zoning is the D-UP and D-DN damper pair on hydronic sheet M-5.
- Zoning: the hydronic pack runs the air side from T4 downstairs, with one damper (D-UP) for this floor (sheet M-5). Keep the two air zones drawn here, which adds T6, T7 and a zone panel to the M-5 schedule, or run this floor as one zone on D-UP with balancing dampers only?

- Radiant floor (hydronic sheet M-1): PEX runs down and back in every truss bay 2" below the subfloor, R-19 batts under it, and the loop leaders run along the west wall through truss web holes. Every register boot and branch here passes through that layer, and the five west-wall registers (G1, G3, G4, G12, G21) land where the leaders and loop ends run. Lay the registers out between PEX passes before the tubing goes in, and give the truss maker the leader holes with the duct routes.
- Truss coordination: the 16" × 8" trunk and the 4" drain at X 40' cross at Y 11'; the truss manufacturer needs both routes.
- Whether the deck-side (west) registers can sit under the windows as drawn or should move to the toe-kick of the island and sofas.

Schedules

Registers and returns

Tag	Room	Position	Grille	CFM	Duct	Run
G1	Elevator hall	X 14'-0", Y 1'-10"	4×10	45	5"	9'-2"
G2	Office	X 8'-6", Y 17'-6"	4×12	65	6"	6'-6"
G3	Living, west	X 20'-0", Y 1'-10"	4×12	90	6"	9'-2"
G4	Living, west	X 34'-0", Y 1'-10"	4×12	90	6"	9'-2"
G5	Powder / hall	X 4'-0", Y 22'-6"	4×10	40	5"	11'-10"
G6	Master bath	X 7'-0", Y 42'-0"	4×10	60	5"	17'-0"
G7	Master bedroom	X 14'-6", Y 46'-0"	4×12	75	6"	11'-6"
G8	Master bedroom	X 26'-6", Y 46'-0"	4×12	75	6"	4'-6"
G9	Walk-in	X 12'-0", Y 27'-0"	4×10	30	5"	12'-0"
G10	Dining / east entry	X 33'-0", Y 46'-0"	4×12	90	6"	11'-0"
G11	Kitchen	X 44'-6", Y 24'-6"	4×12	65	6"	2'-6"
G12	Pantry	X 50'-6", Y 2'-0"	4×10	30	5"	9'-0"
G13	Laundry	X 43'-0", Y 30'-6"	4×10	45	5"	4'-0"
G14	Suite bath	X 51'-6", Y 31'-6"	4×10	45	5"	4'-6"
G15	Den	X 41'-0", Y 46'-0"	4×12	90	6"	8'-0"
G16	2nd Suite	X 56'-6", Y 46'-0"	4×12	75	6"	4'-0"
G17	2nd Suite	X 66'-0", Y 42'-0"	4×10	60	5"	9'-6"
G18	Bedroom 4	X 66'-0", Y 28'-6"	4×12	75	6"	9'-6"
G19	Jack & Jill	X 63'-0", Y 15'-0"	4×10	45	5"	6'-6"
G20	Bedroom 3	X 66'-0", Y 10'-6"	4×12	75	6"	9'-6"
G21	Bedroom 3	X 57'-0", Y 1'-10"	4×10	60	5"	9'-8"
R1	Living, wall 27	X 16'-6", Y 8'-0"	2 bays, 2 × 14×8 wall	390	12×8	27'-1"

Tag	Room	Position	Grille	CFM	Duct	Run
R2	Master bedroom, wall 19	X 28'-4", Y 34'-0"	1 bays, 1 × 14×8 wall	190	8×8	48'-1"
R3	Wing hall, wall 7	X 54'-8", Y 21'-0"	2 bays, 2 × 14×8 wall	310	10×8	43'-2"
R4	Den, wall 17	X 47'-6", Y 36'-8"	1 bays, 1 × 14×8 wall	220	8×8	22'-10"
R5	Dining, wall 32	X 41'-0", Y 33'-0"	1 bays, 1 × 14×8 wall	220	8×8	41'-6"

Supply trunks

Trunk	Size	Length	Route
ST1	18×10	30'-5"	main trunk, 1,325 CFM (this floor's share), east under the passage, office and living
ST2	16×8	26'-6"	665 CFM on, under the kitchen and the wing hall to the bedrooms
ST3	10×8	33'-0"	330 CFM north to the master suite and the east entry
ST4	8×8	33'-0"	245 CFM to the kitchen, laundry, suite bath and den
ST5	10×8	31'-0"	390 CFM under Bedrooms 3 and 4 and the 2nd Suite

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
Rectangular supply trunk 18×10, galvanised, R-6 wrapped	30 lf	plus 10% for fittings
Rectangular supply trunk 16×8, galvanised, R-6 wrapped	26 lf	plus 10% for fittings
Rectangular supply trunk 10×8, galvanised, R-6 wrapped	64 lf	plus 10% for fittings
Rectangular supply trunk 8×8, galvanised, R-6 wrapped	33 lf	plus 10% for fittings
Rectangular return duct 12×8, galvanised	27 lf	
Rectangular return duct 10×8, galvanised	43 lf	
Rectangular return duct 8×8, galvanised	112 lf	
Round branch 5", insulated flex or snap-lock	123 lf	10 runs, 3' each extra for the boot
Round branch 6", insulated flex or snap-lock	118 lf	11 runs, 3' each extra for the boot
Floor register 4×10 with damper, flush steel	10	with floor boot
Floor register 4×12 with damper, flush steel	11	with floor boot
Wall return grilles 14×8 at the baseboard, one per stud bay	7	5 returns; lined bays joined by plenum boxes in the truss space

Item	Qty	Notes
Take-off collars with dampers	21	one per branch
Zone dampers, motorised, with a 2-zone control panel	2	on ST3/ST4 and ST5; see the zoning open item
Thermostats	2	T6, T7, 48" AFF; see the zoning open item
Supply riser boot 18×10 and return riser boot 20×20 through the floor	1 each	at X 4'-4"/Y 6'-4" and X 5'-8"/Y 7'-8"; coordinate the floor opening with the truss layout
Mastic, tape, hangers	lot	

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.